AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M269
Algorithms, Data Structures and Computability

Mid-Term Examination (MTA) – Makeup
Fall Semester 2025/2026

Date: 4/Dec./2025

Number of exam pages (4) (including this cover sheet)	Time Allowed: 2 Hours

Instructions:

- Total Marks: 60
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted
- Be sure you write your name and ID on the External Answer booklet.

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

For the following statements, please answer "True" or "False":

1) Finding the largest element in a queue of size M is of O(M log M).
2) The queue is best data-structure in case the given data is served last in last out.
3) Sorting a Heap can be implemented using Max-Heapify algorithm.
4) The doubly linked list uses less space than singly linked list.
5) Merge sort algorithm uses more memory than the Heap sort Algorithm.
6) When determining the efficiency of an algorithm, the space factor is measured by the maximum memory needed
7) The complexity of a program depends on the lines of code
8) Sorting an ordered list using insertion sort algorithm is better than doing it using merge sort algorithm.
9) $2n^2$ is less in complexity than 200n
10) The selection-sort performs more swapping steps in comparison with the bubble-sort algorithm.

PART 2: ALL QUESTIONS ARE REQUIRED [20 Marks]

Question 2: (10 marks)

a) What is the complexity of the following tasks? (4 marks)
   1) Serve the last customer in a queue of size N.
   2) Sorting a list of N different elements using Merge-Sort algorithm.
   3) Pop the first item entered in a stack data-structure of size N.
   4) Sorting a list of N identical elements using Bubble-Sort algorithm.
   5) Computing the average of N items in a linked list

b) For each of the following cases select the "best" data structure: (4 marks)
   1) A browser needs to remember last visited websites.
   2) Representing the relation between relatives in a family.
   3) A grocery store serving customers based on "come first served first".
   4) A structure that has flexible memory management and its data change frequently
   5) Managing a list of tasks arriving at a processor serving the last task first

Question 3: (10 marks)

Consider the following array => [40, 7, 60, 65, 15, 80, 30, 90, -50, 20]

Each of the following is a view of a sort in progress of the above array. Which sort is which? Each sort is used exactly once. (2 marks each)

Consider the following guides

- Choose between **bubble** sort, **selection** sort, **insertion** sort, **merge** sort, and **quick** sort.
- If the sorting algorithm is merge sort, the array is shown after the recursive calls have completed on each sub-part of the array.
- If the sorting algorithm is quick sort, the algorithm chooses the *first* element as its pivot. For quick sort, the array is shown before the recursive calls are made.

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a) [-50, 7, 15, 65, 60, 80, 30, 90, 40, 20] Sorting Algorithm is .....
b) [7, 20, -50, 30, 15, 40, 65, 90, 60, 80] Sorting Algorithm is .....
c) [7, 40, 60, 65, 15, 80, 30, 90, -50, 20] Sorting Algorithm is .....
d) [7, 15, 40, 60, 65, -50, 20, 30, 80, 90] Sorting Algorithm is .....
e) [7, 15, 40, 30, 60, -50, 20, 65, 80, 90] Sorting Algorithm is .....

PART 3: ALL QUESTIONS ARE REQUIRED [30 marks]

Question 4: (10 marks)

a) Find the complexity of the following blocks of code, where Arr is an array, n is the input size:
[Note: your answer must show the steps that lead to your final answer] [2 marks each]

i)
```
for i = 1 to n do
    for k = 1 to n do
        for (j = 2 ; j < n ; j *= 2)
            Arr[i] = i + k + j;
        end for
    end for
end for
```

ii)
```
val = 1
for k = 1 to 10
    val = val × k
end for
for l = 1 to 10
    Arr[l] = Arr[l] + val
end for
```

iii)
```
for i = 1 to 100 do
    for k = 1 to n do
        j = 1 ; m = n;
        while j <  m do
            m = m / 2;
        end while
    end for
end for
```

b) Solve the following linear recurrences and find the complexity: [2 marks each]
i) $T(n) = T(n-1) + 5$ for $n > 1$ with $T(1) = 0$
ii) $T(n) = T(n-1) + n$ for $n > 0$ with $T(0) = 0$

Question 5: (10 marks)

a) Consider the Uniqueness Test Algorithm:
Check whether all the elements in a given array are distinct
- Input: An array A[0...n-1]
- Output: Return "true" if all the elements in A are distinct and "false" otherwise

```
Algorithm UniqueElements ( A[0 ... n-1])
for i ← 0 to n-2 do
    for j ← i+1 to n-1 do
        if A[i] = A[j] return false
```


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1) Analyze the complexity of this algorithm. (2 marks)
2) Propose another algorithm to do the job with less complexity (3 marks)

b) Given the following array A [7,12,4,9,19,2,6,8,11,15,20,5,3], answer the following set of questions:
1) Draw the heap presentation of this array (1 mark)
2) What is the depth of node 19? (1 mark)
3) Find the sibling(s) node(s) of node 20. (1 mark)
4) Find the pre-order traversal of this tree. (2 marks)

Question 6: (10 marks)

Given an array of unsorted integers, find the maximum product of two integers in an array. For example, if the arr = [8, 3, 10, 4, 9] then the maximum product would be = 90, as 10 and 9 are the largest integers in this given array and hence their product will be the answer. Answer the following questions:

a) Design an algorithm with the least possible complexity to solve this problem. (6 marks) [Hint: consider unpredicted input]
b) Analyze the complexity of your algorithm. (4 marks)

End of questions

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